6. Work, Power & Energy.

- Work - It is a scalar product of force and displacement.

$$W = \vec{F} \cdot \vec{S} = |F||S|\cos\theta$$

Unit = Nm → Joule.

- When work done is positive then the system gets energy.
- When work done is negative then the system loses energy.

* For variable force

$$W = \int_{x_1}^{x_2} F\,dx$$

* The area of force & displacement gives work done in graph

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* It is not necessary that if displacement is zero, total work done is also zero.

* For forces in multiple directions

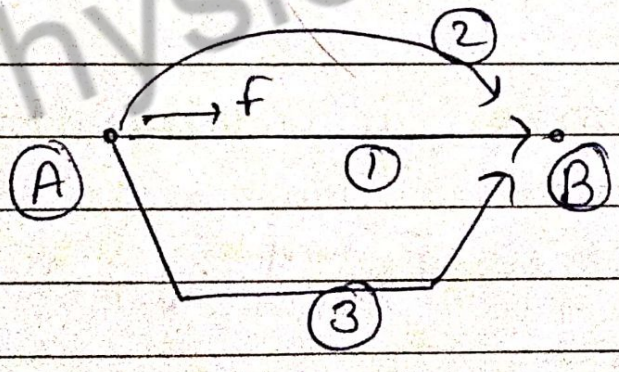
$$W = \int F \, d\vec{r}$$

where $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$.

* Conservative force

- The work done by such force

1. does not depend upon path
2. Depends only & only on initial & final position.



$W_1 = W_2 = W_3$

*) Ex - Gravitational force, spring force, Electrostatic force.

* Work done by conservative force in a closed path is always zero.

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* Non conservative forces are forces which are not conservative.

* Conservative force	Non conservative force
1) $W_I = W_{II}$	$W_I \neq W_{II}$
° does not depend upon path	• does depend on path
• Closed path $W = 0$	Closed path $W \neq 0$

(*) If a Force is conservative

then $F = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$

Check i) $\frac{\delta F_x}{\delta y} = \frac{\delta f_y}{\delta x}$

ii) $\frac{\delta F_x}{\delta z} = \frac{\delta f_z}{\delta x}$

iii) $\frac{\delta f_y}{\delta z} = \frac{\delta F_z}{\delta y}$

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Q) Find whether given force $x\hat{i} + y\hat{j}$ is conservative or non-conservative..

(1) $\frac{dF_x}{dy} = \frac{dF_y}{dx} \Rightarrow 0 = 0$

[F_x will be constant w.r.t y]

[F_y will be constant w.r.t x]

(2) $\frac{dF_y}{dz} = \frac{dF_z}{dy} \Rightarrow 0 = 0$

(3) $\frac{dF_x}{dz} = \frac{dF_z}{dx} \Rightarrow 0 = 0$

$\Rightarrow$ **Given force is conservative.**

* Work Energy Theorem.

- Kinetic energy comes into action when body is in motion.
- It depends upon frame of reference.

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• Work energy theorem.

⇒ Work done by all the forces on a body = Change in K.E. of the body.

↳ internal + external force.

$$W_{\text{by all force}} = K.E._f - K.E._i$$

$$dW_{\text{all forces}} = d(K.E.)$$

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★ Potential Energy

$$\Delta U = -\text{Work done by Conservative force.}$$

$$\Delta U = -W_c$$

$$dU = -dW_c$$

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* Potential energy is not defined for non-conservative force.

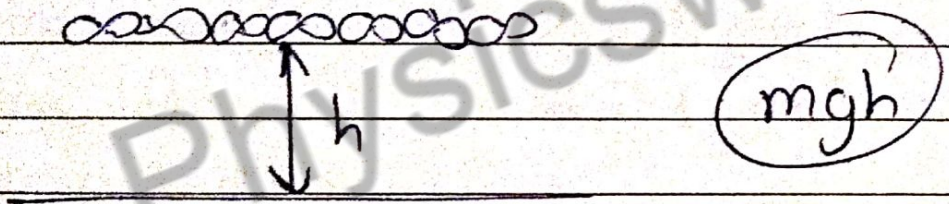
$$\int dU = -\int dW_c$$

$$\int_{U_i}^{U_f} dU = -\int F_c \cdot dx$$

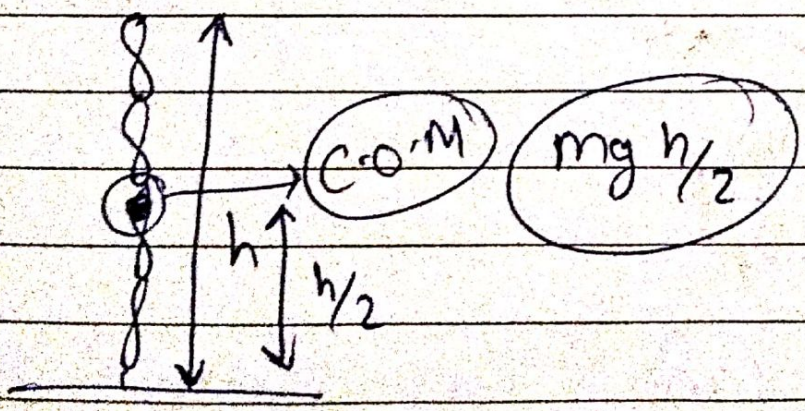
* Gravitational Potential Energy. (mgh)

• Check for Center of mass of for h

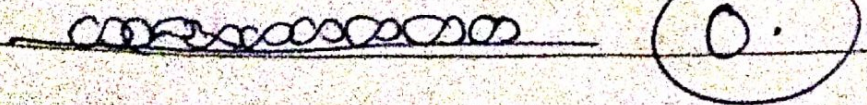
(i)



(ii)



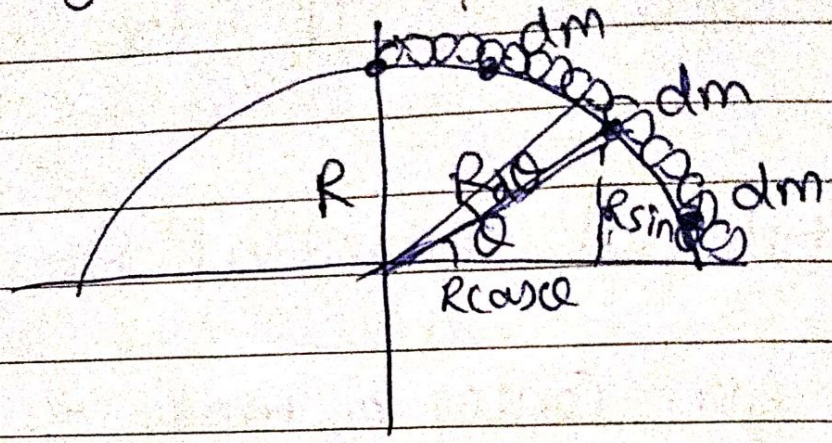
(iii)



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* To find Potential Energy of given arrangement of chain.

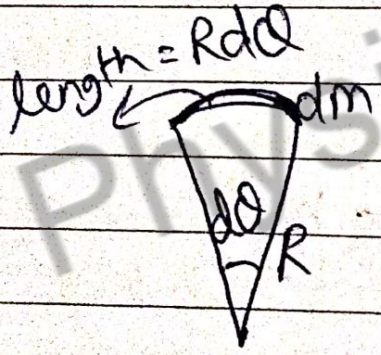


$dU = dmgh$

$\Rightarrow dU = dmgR\sin\theta$

$\Rightarrow \int dU = \int dmgR\sin\theta$

[Here variable is 'θ' not dm, hence make θ variable]



Mass (m) for length $\frac{\pi R}{2}$

then mass ? for length $\int Rd\theta$

$\Rightarrow \frac{mRd\theta}{\pi R} = ?$

$dm \rightarrow \frac{2md\theta}{\pi}$

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* Potential Energy is invisible form of energy it is imparted to the system, that energy is stored as potential energy.

$$\int dU = \int_0^{\pi/2} \frac{2m}{\pi} dQ\, g R \sin\theta$$

$$= \frac{2m}{\pi} gR \left[\sin\theta\right]_0^{\frac{\pi}{2}}$$

$$\Rightarrow \frac{2m}{\pi} gR \left[-\cos\theta\right]_0^{\frac{\pi}{2}}$$

$$\Rightarrow \frac{2m}{\pi} gR \left[-\cos 90 - (-\cos 0)\right]$$

$$\Rightarrow \boxed{\frac{2mgR}{\pi}}$$

* Spring Potential Energy.

$$U = \frac{1}{2} K x^2$$

° Work done by spring force

$$= \frac{1}{2} K (x_1^2 - x_2^2)$$

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* Potential Energy of spring is zero at its natural length.

(*) $U_2 - U_1 = \frac{1}{2}Kx_2^2 - \frac{1}{2}Kx_1^2$

- Either you have stretched x, or compressed x ~~for~~ from natural length the potential energy stored will always be equal to $\frac{1}{2}Kx^2$.

- Potential Energy of spring is always positive.

- Gravitational & Electrostatic potential energy can be negative.

(*) $F_x = -\frac{dU}{dx}$

(*) $F_y = -\frac{dU}{dy}$

(*) $F_z = -\frac{dU}{dz}$

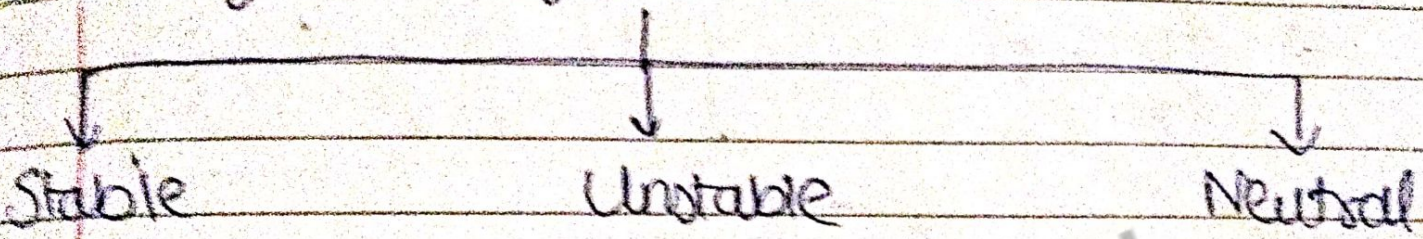
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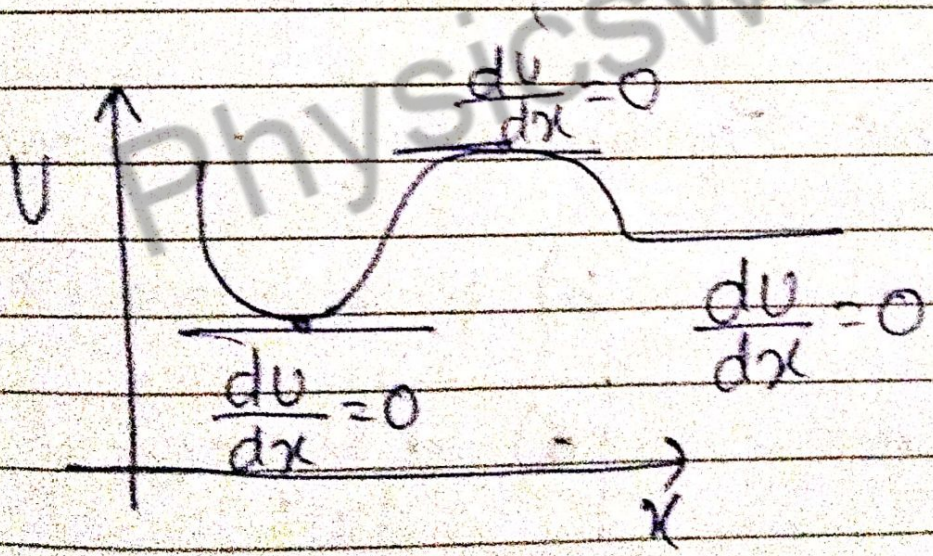
(*) Slope of U vs x graph is Conservative force.

* Equilibrium.

- If total force acting on body is zero, then body is in Equilibrium.



(*) F_x = -ve (Slope U v/s x)



(*) Stable Equilibrium ⇒ U → minimum

(*) Unstable Equilibrium ⇒ U → maximum

(*) Neutral Equilibrium ⇒ U → Constant.

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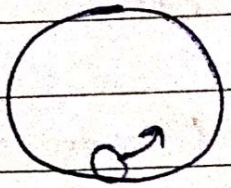
(1.) Stable Equilibrium [U = minimum]

a.) $F_{net} = 0$

b.) U - x slope = 0

c) $\frac{dU}{dx} = 0$

d) $\frac{d^2U}{dx^2} > 0$



If we displace the body slightly it, returns back to stable equilibrium.

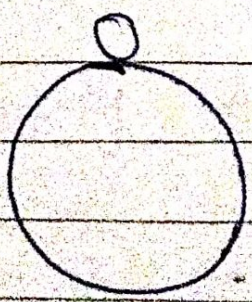
(2.) Unstable Equilibrium [U - maximum]

a) $F_{net} = 0$

b) U - x slope = 0

c) $\frac{dU}{dx} = 0$

d) $\frac{d^2U}{dx^2} < 0$



If we displace the body slightly it will move towards ~~stable~~ stable equilibrium.

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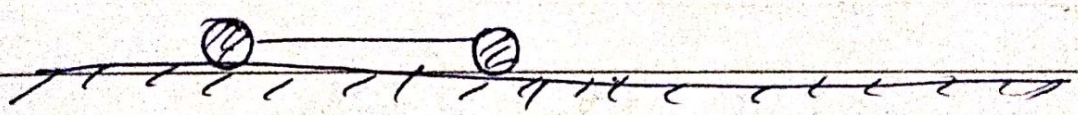
③ Neutral Equilibrium [U - constant]

a) $F_{net} = 0$

b) U - x slope = 0

c) $\frac{dU}{dx} = 0$

d) $\frac{d^2U}{dx^2} = 0$



will remain in again in neutral equilibrium position [new position]

★ Conservation of Mechanical Energy

- $W_e + W_{N.C.} = (K_f + U_f) - (K_i + U_i)$

⇒ When external force = 0
Non - conservative force = 0

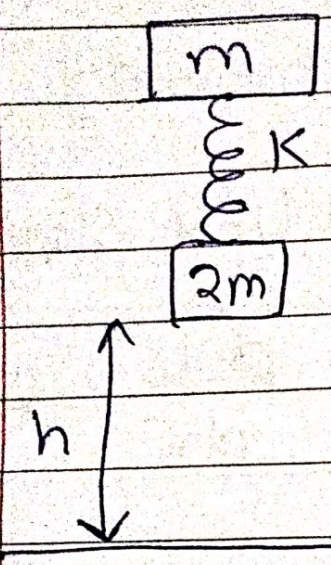
then, $K_i + U_i = K_f + U_f$

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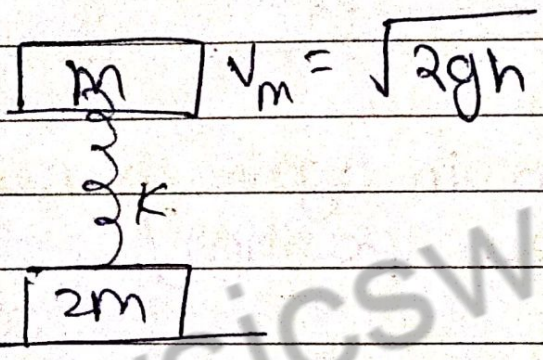
P6
RDV

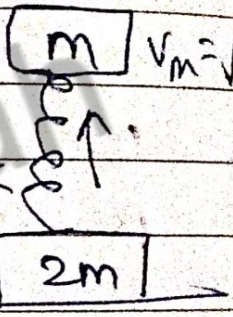
Que

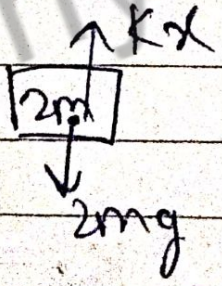


Find the value of h such that 2m just manages to rise above ground after collision.

⇒





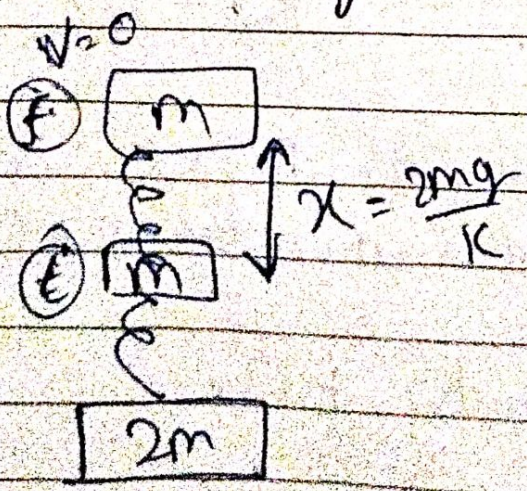


⇒ Kx should just more than 2mg

⇒ We can take

$$Kx = 2mg$$

$$x = \frac{2mg}{K}$$



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Using Conservation of mechanical Energy

$$K_f + U_f = K_i + U_i$$

$$0 + mgx + \frac{1}{2}kx^2 = \frac{1}{2}m \cdot 2gh + 0$$

$$\Rightarrow mgx + \frac{1}{2}kx^2 = \frac{1}{2} \cdot 2mgh$$

$$\Rightarrow \frac{2mgx + kx^2}{2} = mgh$$

$$\Rightarrow 2mgx + kx^2 = 2mgh$$

$$\Rightarrow kx^2 + 2mgx - 2mgh = 0$$

$$x = \frac{2mg}{k}$$

$$\Rightarrow k \cdot \frac{4m^2g^2}{k^2} + 2mg\,\frac{2mg}{k} - 2mgh = 0$$

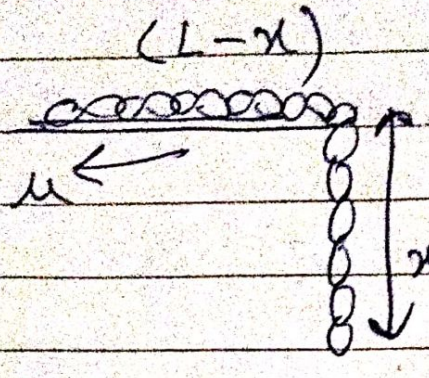
$$\Rightarrow \frac{4m^2g^2}{k} + \frac{4m^2g^2}{k} = 2mgh$$

$$\Rightarrow \frac{8m^2g^2}{2k} = 2mgh$$

$$\Rightarrow 4m^2g^2 = mgh \Rightarrow \boxed{h = \frac{4mg}{k}}$$

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Que 

A chain of length 'L' and mass 'M' is made to sustain on the table find the maximum length it can hang so that body doesn't falls.

⇒ Let the chain to be hanged be of length x.

∴ chain on table = (L-x)

If L length of chain weighs M
the x length of chain weights $\frac{Mx}{L}$

As (L-x) length of chain weight $\frac{M(L-x)}{L}$

Now frictional force = μN

$\Rightarrow \mu(mg)$

$\Rightarrow \mu \frac{M(L-x)}{L} g$

Force acting on falling chain = $\frac{Mx}{L} g$

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⇒ As body has to stay

$$\therefore \frac{\mu M(L-x)g}{L} = \frac{Mxg}{L}$$

$$\Rightarrow \mu(L-x) = x$$

$$\Rightarrow \mu L - \mu x = x$$

$$\Rightarrow \mu L = x + \mu x$$

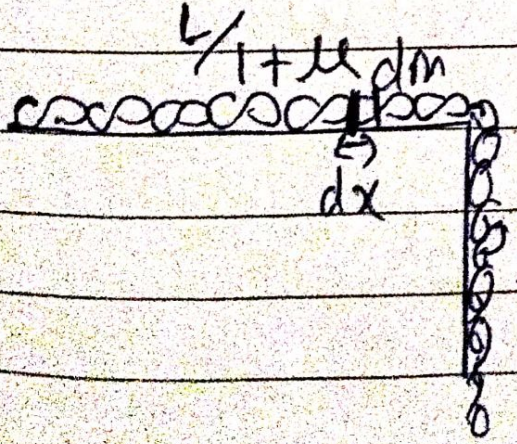
$$\Rightarrow \mu L = x(1+\mu)$$

$$\Rightarrow \boxed{\frac{\mu L}{1+\mu} = x}$$

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Que On the same Que above find the total work done by frictional force if the chain is pulled horizontal down.

⇒



$\frac{\mu L}{1+\mu}$

If $L \to M$

$dx \to \frac{M dx}{L}$

$$f = \mu mg = \mu \frac{M dx}{L} g$$

$$dw = fx$$

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$$\int dw = \int \frac{\mu M dx g x}{L}$$

$$\Rightarrow \int dw = \frac{\mu M g}{L} \int x\, dx$$

Vary x from $\frac{L}{1+\mu}$ to 0.

$$\therefore \int dw = \frac{\mu M g}{L} \int_0^{\frac{L}{1+\mu}} x\, dx$$

$$\Rightarrow \frac{\mu M g}{L} \left[\frac{x^2}{2}\right]_0^{\frac{L}{1+\mu}}$$

$$= \frac{\mu M g}{2L} \left[\left(\frac{L}{1+\mu}\right)^2\right]$$

$$\Rightarrow \frac{\mu M g L^2}{2L(1+\mu)^2}$$

$$\Rightarrow \frac{\mu M g L}{2(1+\mu)^2}.$$

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